FIFA WORLD CUP 2026: DATABASE DESIGN AND

SECURITY IMPLEMENTATION

DATABASE 281

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Business Rules

Introduction

The 2026 FIFA World Cup database must adhere to a set of Business Rules to function properly and accurately. These regulations serve as the system's logic, ensuring that information is accurately collected and maintained throughout the competition in the USA, Mexico, and Canada, including match scores, player rosters, and ticket sales.

1. **Stadium Locations**: Each stadium must be situated in one of the three host nations' cities.
2. **National Teams**: Every team must have a distinct official FIFA identification code and represent a single nation.
3. **Player Registration:** During the competition, a player may only represent one national team.
4. **Coaches:** A head coach is required for each team. Several assistant coaches might also assist them.
5. **Match Setup:** A team cannot play against itself; all scheduled matches must be between two distinct teams.
6. **Venues:** Only one stadium may host a match at a time.
7. **Schedules**: Each game needs to have a set kick-off time and a confirmed date.
8. **Tournament Rounds:** Each match needs to be identified by its stage, such as Group Stage or Final.
9. **Purchasing Tickets:** A fan may purchase multiple tickets, but each ticket must be for a certain seat at a particular game.
10. **Seat Numbers:** No two seats in the same stadium may share the same Section, Row, and Seat number.
11. **Final Scores**: For each game, the database needs to keep track of how many goals each team scored.
12. **Full Capacity:** The system will not permit the sale of more tickets for a game than the stadium's total number of seats.

Database Normalization (UNF to 3NF)

Unnormalized Form (UNF)

In this stage, the data is unorganized. A single row contains multiple players and repeating information.

MatchID	Teams	Stadium	Players
M101	USA vs Mexico	Azteca	P. Pulisic, G. Reyna, H. Lozano

Problem: It is hard to search for a single player because the Players column has numerous values in a single cell (repeating groupings).

First Normal Form (1NF)

To reach 1NF, we ensure every cell contains only one value and every row is unique.

MatchID	TeamName	StadiumName	PlayerName
M101	USA	Azteca	P. Pulisic
M101	USA	Azteca	G. Reyna
M101	MEXICO	Azteca	H. Lozano

Action Taken: Removed repeating groups. Each player now has their own row.

Problem: Azteca appears frequently. We must update hundreds of rows (Redundancy) whenever the stadium name is changed.

Second Normal Form (2NF)

To reach 2NF, we remove Partial Dependencies. We split the data into different tables based on what they describe (Teams, Players, Stadiums)

Teams
PK - TeamID
TeamName

Players
PK - PlayerID
PlayerName
FK - TeamID

Matches
PK - MatchID
StadiumName
MatchDate

Third Normal Form (3NF)

To reach 3NF, we remove Transitive Dependencies. We ensure that no non-key column depends on another non-key column (e.g., a Country depends on a city, not a Stadium).

Final 3NF Table Structure:

Countries: (CountryID, CountryName)

Cities: (CityID, CityName, CountryID)

Stadiums: (StadiumID, StadiumName, Capacity, CityID)

Teams: (TeamID, TeamName, Coach, CountryID)

Players: (PlayerID, PlayerName, Position, TeamID)

Matches: (MatchID, MatchDate, Stage, StadiumID)

ERD Relationships

